Habib University

Course Title: Digital Logic and Design

Instructors: Dr. Aamir Hasan, Ahmed Ali Mustansir, Areeba Aziz Rajput, Dr. Muhammad Farhan

Assignment No. 03

Release Date: 23rd Feb'26	**Due by:** 12th March'26
Total points: 100	**Points obtained:**

Student Name:	**Student ID:**	**Section:**

Purpose:

The purpose of this assignment is to help you practice the number representation; its conversion; and the arithmetic operations in different base formats. In addition, this assignment will aid you to understand the process of representing a signed number in different formats; parity bits; representing complements of a number; and performing BCD arithmetic.

Instructions:

1. This assignment should be done individually.
2. All questions should be answered in **black ink only**. (Extra sheets can be used)
3. Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

1. Your assignments will be checked by instructor/TA.
2. You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to correctly answer the question you have attempted right, you may lose your marks.
3. Zero will be given if the assignment is found to be plagiarized.
4. Untidy work will result in a reduction of your points.

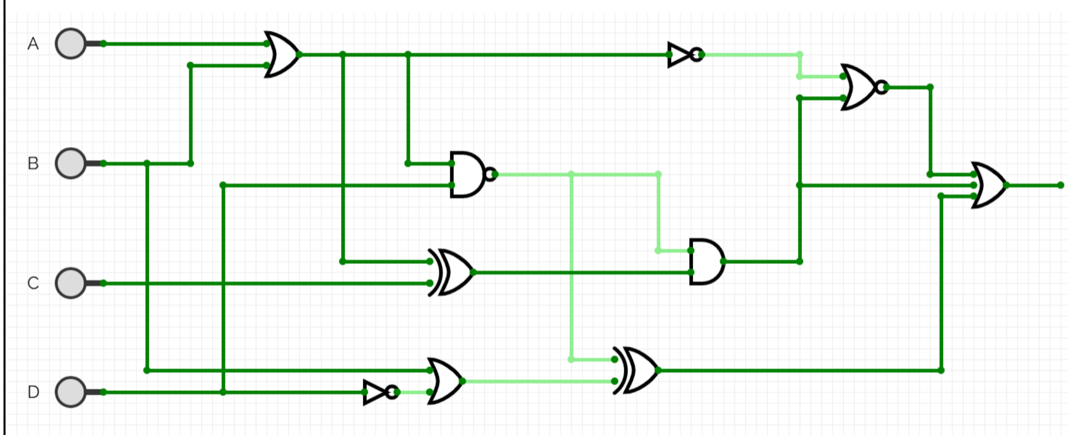
Late submission penalty:

- 1-day late submission – 20% deduction of the maximum allowable marks
- 2-days late submission – 40% deduction of the maximum allowable marks
- No submission will be accepted after two days of the original deadline

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	Apply r-base (binary, octal, decimal, hexadecimal numbers systems) to digital systems and carry out arithmetic operations and conversions	
CLO 2	Apply principles of Boolean Algebra to represent and build equivalent realizations of digital logic (circuits)	
CLO 3	Design combinational logic circuits using logic gates	✓
CLO 4	Design sequential systems using finite state machine methodology	

P#	Questions	Pts
Q1	Design a combinational circuit that generates the 7's complement of an octal digit.	10
Q2	You are required to design a a) 4-to-16 line decoder using four 2-to-4 line decoders with enable inputs. b) 16-to-1 multiplexer using a decoder and logic gates. c) full subtractor using 4x1 demultiplexers.	5x3
Q3	Given the following circuit:  a) Run Analysis procedure i.e. find the Boolean representation of the logic circuit and obtain its truth table. b) Using two 3x8 decoders, implement the expression obtained in part (a) c) Using a 2x1 multiplexer, implement the expression obtained in part (a) d) Using a 8x1 multiplexer, implement the expression obtained in part (a)	10+5+ 5+5
Q4	Design a combinational circuit that generates the 2's complement of a 3-bit binary number. Implement a) One 4x1 multiplexer and necessary gates (if needed). b) One 3-to-8 decoder with enable input. Explain which implementation is more efficient in terms of gate count and why.	6x2+2
Q5	In a pharmaceutical plant, a conveyor belt is carrying 500-mg medicinal tablets for final quality control. Each tablet's weight is measured via a piezoelectric sensor, which generates a number of pulses proportional to the deviation in weight that must be corrected. ● The generated pulses are proportional to the delta in weight, covering deviations from +10 mg to –10 mg. ● To meet regulatory requirements, the negative delta (underweight) must remain within 8 mg. To ensure this, the filler is activated.	7x2+6

- For financial feasibility, the maximum surplus per tablet must not exceed 5 mg. Therefore, a micro-drilling nozzle is activated to remove excess coating material if the tablet is overweight.
- The pulses are fed into a 3-bit counter. The counter's outputs (A, B, C, with A as the MSB) are connected to a logic circuit that drives the conveyor belt's motor control signal, X.
- The motor is designed to pause whenever the correction system (nozzle or filler) is active, ensuring each tablet's weight deviation is mitigated before continuing down the line

a) Fill out the table below and provide explanation of your logic while filling the table

Weight delta (mg)	A	B	C	X (motor brakes)	Y (nozzle)	Z (filler)
+10	0	0	0			
+7.5	0	0	1			
+5	0	1	0			
+2.5	0	1	1			
-2.5	1	0	0			
0	1	0	1			
-5	1	1	0			
-10	1	1	1			

b) Design the new Logic to control the brakes, X, the mechanism controls nozzle function, Y and the filler function, Z. Make your best judgement to either use K-MAP for POS minimization or SOP minimization

c) If the MSB of counter output is grounded and the motor is controlled by the last 2-bits. Redraw another table and design another logic circuit. Conclude on your findings.

Q6	Your company, HUIC Technologies, recently hired a junior auraless IC designer, Krish, and assigned him the task of developing a new decimal-to-BCD encoder. As his supervisor, you were asked to review his progress. Upon receiving the truth table Krish prepared, you were shocked at the level of unprofessionalism and noticed that the	8x2

design contained a serious fault. The encoder was producing ambiguous outputs when multiple decimal inputs were active at the same time, which is unacceptable for a commercial IC.

I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9	I_{10}	A	B	C	D
0	0	0	0	0	0	0	0	1	0	1	0	0	0
0	0	0	0	0	1	0	0	0	0	0	1	0	1
0	0	1	0	0	0	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	0	0	0	1	0	0
0	1	0	0	0	0	0	0	0	0	0	0	0	1
0	0	0	0	0	0	1	0	0	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	1	1	1
1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	1	1	0	0	1

a) The truth table received was out of order. You first correct the truth table ordering and then proceed to write the 4 equations in terms of inputs and make the logic circuits (you can draw the circuit on paper or on any software.)

b) Once you have the equations and the logic circuit you proceed to find the fault in the design and propose your own design and implementation of it. (Hence make the new truth table, and equations for your design).

Good job! You saved your company from millions of dollars in loss, and not to mention the humiliation that you and your team would have had to face if Krish’s unprofessionalism went unnoticed.